

LUMINA ENERGY · HELIOS PLATFORM

Product Vision

The operating system for the energy transition

Owner: Meera Pillai, VP Product | Status: Active — reviewed annually at Q4 planning | Last Updated: 2024-10-02

"We built Helios because the grid is the most critical piece of infrastructure in modern civilization, and it is still largely operated with tools designed in the 1990s. We are not building a dashboard. We are building the operating system for the energy transition."

— James Whitfield, Co-Founder & CTO, Lumina Energy (2020 Series A investor deck)

North Star

NORTH STAR Helios exists to make national electrical grids safer, more reliable, and more sustainable by giving grid operators the intelligence they need to act faster than failures can propagate.

This statement was finalized in Q3 2021 after our first real production incident: a cascade failure at a utility customer that our system saw coming four minutes before it happened but could not surface to operators fast enough to act on. That incident fundamentally shaped every major architectural decision since.

The Problem We're Solving

The Grid Is Getting Harder to Operate

For most of the 20th century, electrical grids were relatively predictable systems. Power flowed in one direction — from large centralized generators downstream to consumers. That world is gone. Modern grids must now contend with:

- Distributed generation — solar on millions of rooftops producing power flowing in both directions through networks designed for one-way flow
- Intermittency — wind and solar generation that can drop or surge in minutes, creating imbalances inconceivable at scale with legacy tools
- Battery storage — grid-scale and residential batteries adding new degrees of freedom SCADA was not designed to handle
- EV adoption — millions of electric vehicles creating enormous, poorly predictable demand spikes during evening charging windows
- Climate events — more frequent extreme weather stressing infrastructure precisely when demand is highest
- Aging infrastructure — most of the physical grid in the US, Europe, and Australia was built between 1950 and 1980

The Status Quo Is SCADA

The incumbent technology for grid management is SCADA — Supervisory Control and Data Acquisition. SCADA systems are operationally reliable but architecturally frozen. They were designed for polling-based telemetry at low frequency, not streaming telemetry from tens of millions of IoT endpoints. They have no native AI capability. Their UIs are often 15–20 years old. Our customers are not replacing SCADA — they are keeping it for low-level control functions it does reliably, and deploying Helios as the intelligence and operations layer on top of it.

Vision Pillars

01 Operational Awareness at Grid Scale Every asset visible in a single interface, updated in near real-time. Target: ≤4s P95 end-to-end latency. Current (Q4 2024): 3.7s P95.	02 Predictive Intelligence AI-driven forecasting of demand, outage probability, and equipment health. Operators act on what’s about to happen. Current horizon: 7 days at 15-min resolution.
03 Rapid Outage Response From outage detection to technician dispatch in under 4 minutes. Automated severity escalation and intelligent routing minimize SAIDI and SAIFI.	04 Regulatory Confidence Pre-formatted OFGEM, BDEW, and AEMO reporting. Immutable 7-year audit logs. GDPR-compliant data handling. Compliance built in, not bolted on.

Where We Are Now — Q4 2024

14 Utility Customers	42M Smart Meters	6 Countries	99.961 % Platform Uptime	71 Engineers	v4.7.1 Current Version
--	--	-----------------------------------	---	------------------------------------	--

What Helios Is Not

NOT THIS	RATIONALE
A SCADA replacement	We complement SCADA. We don't replace low-level control systems — that's a different safe
A consumer energy app	We serve utility companies, not end consumers. A consumer app would compete with our cu
A hardware company	We integrate with hardware vendors. We don't manufacture devices.
A billing platform	Several customers asked. We integrate with billing via API. Billing is not our domain.
Real-time automated switching (now)	Requires IEC 61850 safety certification. On the long-term roadmap, not current scope.

Strategic Priorities 2025

- US Market Entry (Q3 2025) — First US customers, NERC CIP compliance foundation, SOC 2 Type II certification.
- Virtual Power Plant Module (Q2 2025) — Schedule-based battery optimization for ENGIE and NW Grid UK VPP programs. €2M contract extension in scope for ENGIE.
- Platform Reliability & Security Investment — Complete mTLS rollout, ISO 27001 certification, IoT cross-region DR gap remediation, dispatch service Node 20 modernization.